

L-2-2: Useful Statistical Procedures for Nowcasting, Part 2: Stationarity and ARIMA Models

OT26.07 Cohort Training in Forecasting and Nowcasting
September 21 – October 2, 2026

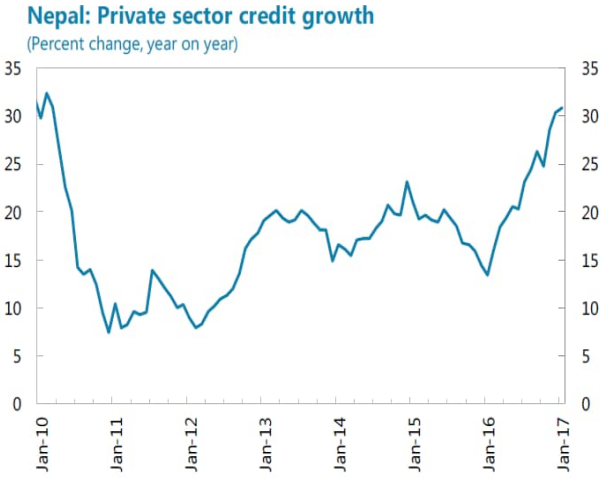
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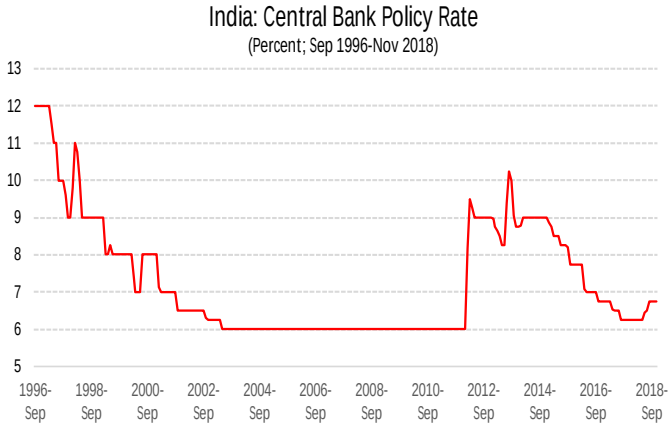
Motivation

- Most data in macroeconomics and finance are a series of data points indexed by time: i.e., ***time series***.
- Time series ***analysis*** *consists of* methods for analyzing time series data to extract meaningful statistics and other characteristics of the data.
- Time series ***forecasting*** is the use of a model to predict future values using previously observed values.

How to model these time series?



Source: Nepal, 2017 IMF Article IV Staff Report



Source: Haver Analytics

Outline

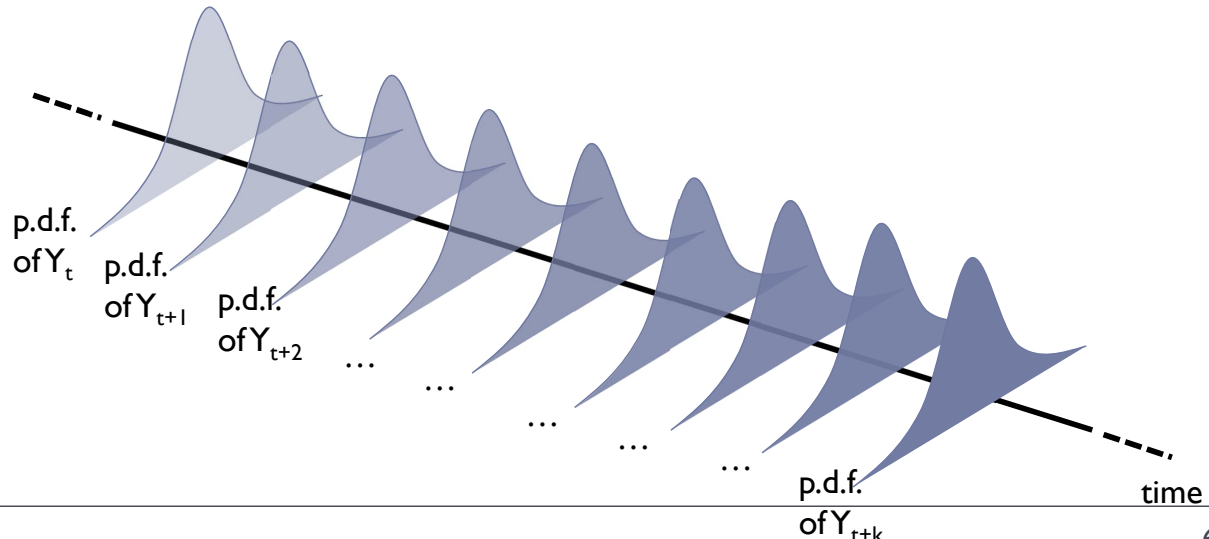
I. Time series and stationary processes
II. ARMA processes
III. Stationarity of ARMA process
IV. Correlogram
V. The Box-Jenkins methodology
VI. Forecasting
VII. Appendixes

Part I. Time series and stationary processes

- Stochastic process
- Time series
- Stationary processes

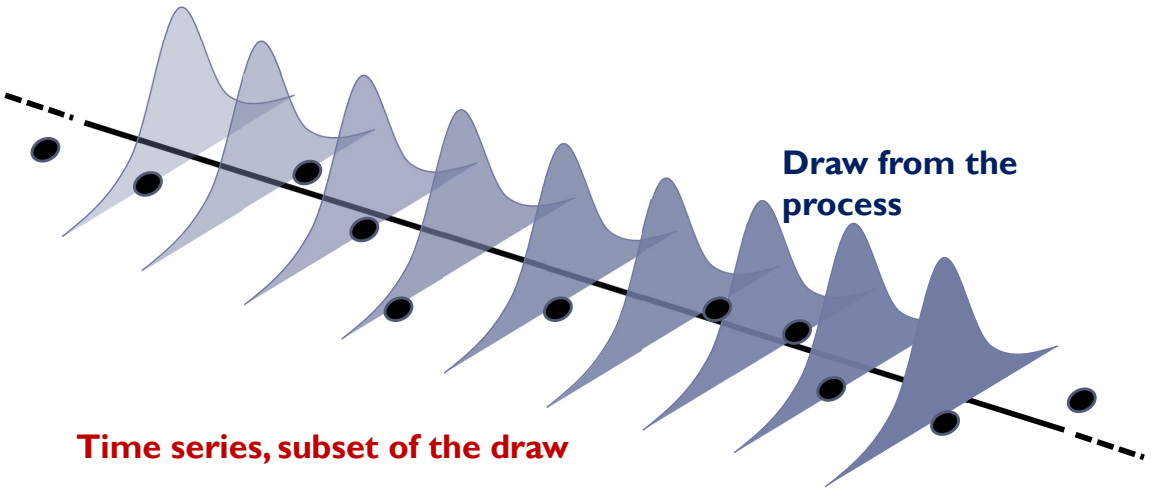
Stochastic Process

A stochastic process is an infinite collection $\{Y_t\}_{-\infty}^{+\infty}$ of random variables Y indexed by time



Time Series

A time series is a subset $\{y_t\}_{t=0}^{T}$ of one draw $\{y_t\}_{-\infty}^{+\infty}$ from the stochastic process $\{Y_t\}_{-\infty}^{+\infty}$



Stationarity

A stochastic process with *finite* mean and variances is (covariance/weakly) stationary if:

- **the unconditional mean and variance are constant over time**

$$E(Y_t) = E(Y_{t+j}) = \mu \qquad Var(Y_t) = Var(Y_{t+j}) = \sigma_Y^2$$

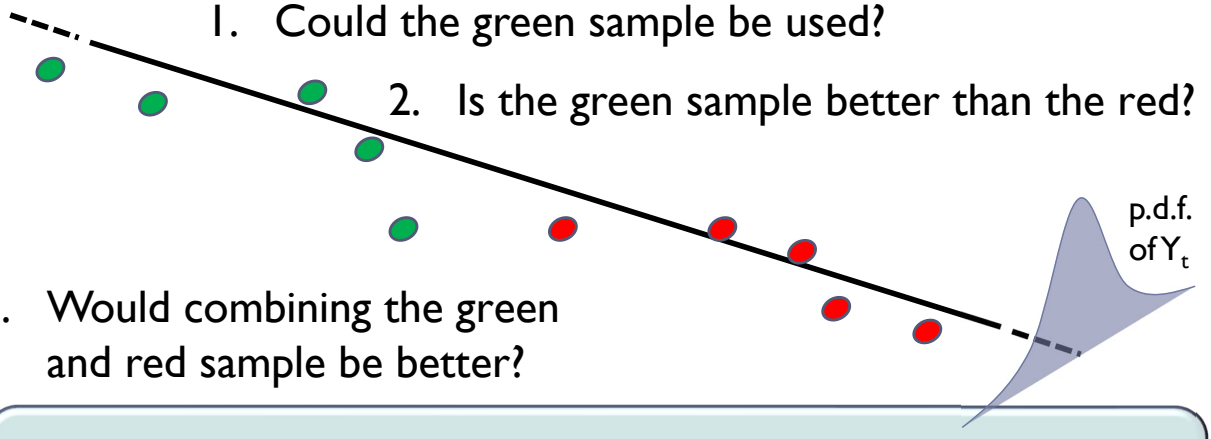
- **the covariance depends only on the time *j* that has lapsed, and not on the reference period *t***

$$Cov(Y_t, Y_{t+j}) = Cov(Y_s, Y_{s+j}) = \gamma_j$$

Importance of Stationarity

Consider a time series from a stationary process. Suppose one wants to make inference on, say, $E(Y_t)$ or $Var(Y_t)$ or $Cov(Y_t, Y_{t+j})$?

1. Could the green sample be used?
2. Is the green sample better than the red?
3. Would combining the green and red sample be better?



If a stochastic process is stationary, then any observed time series suffices to make inference about the properties of the process. The longer the series, the more precise the inference.

Part II. ARMA Processes

- AR processes
- MA processes
- ARMA processes

AR process

A stochastic process admits an AR(p) representation (follows an autoregressive process of order p) if it can be represented as

$$y_t = a + b_1 y_{t-1} + b_2 y_{t-2} + \ldots + b_p y_{t-p} + \varepsilon_t$$

ε_t independent and identically distributed

We will assume:

$$E[\varepsilon_t] = 0$$
$$Var[\varepsilon_t] = E[\varepsilon_t^2] = \sigma^2$$
$$E[\varepsilon_t \varepsilon_{t-j}] = 0$$

(White noise)

MA process

A stochastic process admits an MA(q) representation (follows a moving average process of order q) if it can be represented as

$$y_t = \mu + \varepsilon_t + \phi_1 \varepsilon_{t-1} + \phi_2 \varepsilon_{t-2} + \ldots + \phi_q \varepsilon_{t-q}$$

ε_t independent and identically distributed

We will assume:

$$E[\varepsilon_t] = 0$$
$$Var[\varepsilon_t] = E[\varepsilon_t^2] = \sigma^2$$
$$E[\varepsilon_t \varepsilon_{t-j}] = 0$$

(White noise)

ARMA process

A stochastic process admits an ARMA(p,q) representation (follows an ARMA(p,q) process) if

$$y_t = a + \underbrace{b_1 y_{t-1} + b_2 y_{t-2} + \ldots + b_p y_{t-p}}_{\text{AR(p) component}} + \varepsilon_t + \underbrace{\phi_1 \varepsilon_{t-1} + \phi_2 \varepsilon_{t-2} + \ldots + \phi_q \varepsilon_{t-q}}_{\text{MA(q) component}}$$

ε_t independent and identically distributed

We will assume:

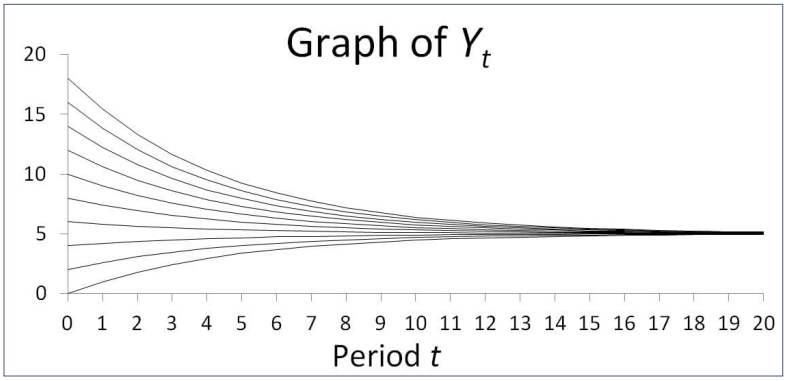
$$E[\varepsilon_t] = 0$$
$$Var[\varepsilon_t] = E[\varepsilon_t^2] = \sigma^2$$
$$E[\varepsilon_t \varepsilon_{t-j}] = 0$$

Part III. Stationarity of ARMA Processes

- Stationarity of an AR(1) process
- Stationarity of an AR(p) process
- Stationarity of a MA(q) process
- Integrated processes, ARIMA model

Deterministic Process

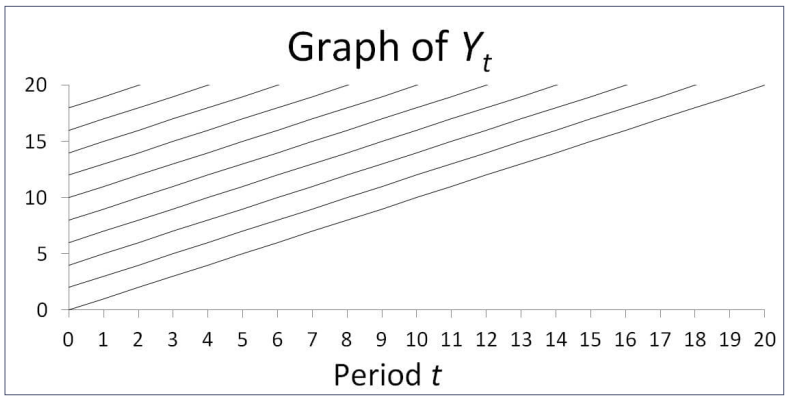
Consider $y_t = 1 + 0.8y_{t-1}$



The variable y converges to a constant for any initial value of (y_0)
(see Appendix 1 for an interpretation of stationarity)

Deterministic Process

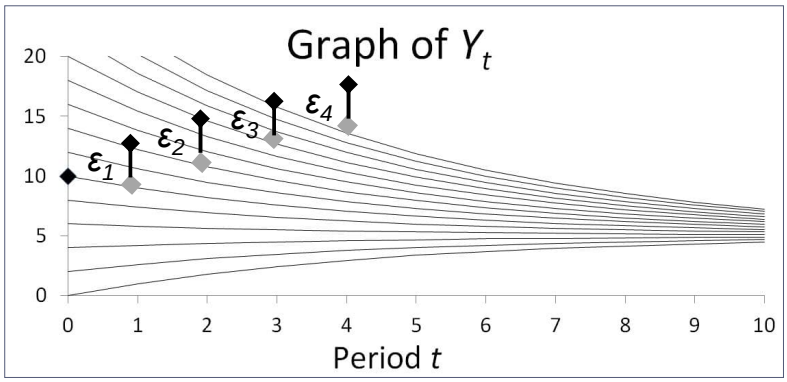
Consider instead $y_t = 1 + y_{t-1}$



For every y_0 (initial value) the variable "explodes" or does not converge to a given value.

Stochastic Process

Consider adding a random component $Y_t = 1 + 0.8Y_{t-1} + \varepsilon_t$



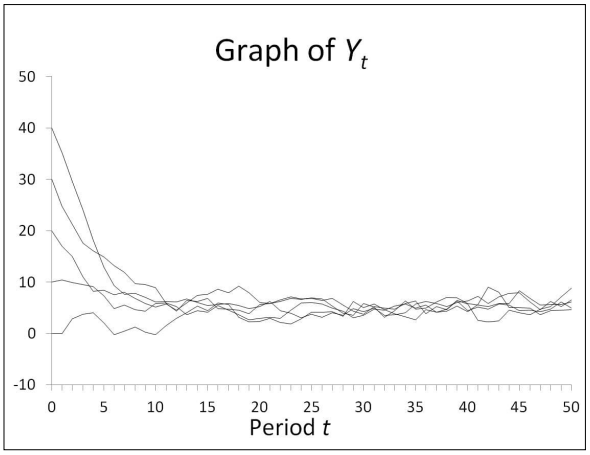
- An explosive behavior is possible, but very unlikely (we would need infinitely many positive and large ε_t)
- Most likely, the disturbances cancel out over time, and the variable Y_t converges (on average) to 5 as time goes by

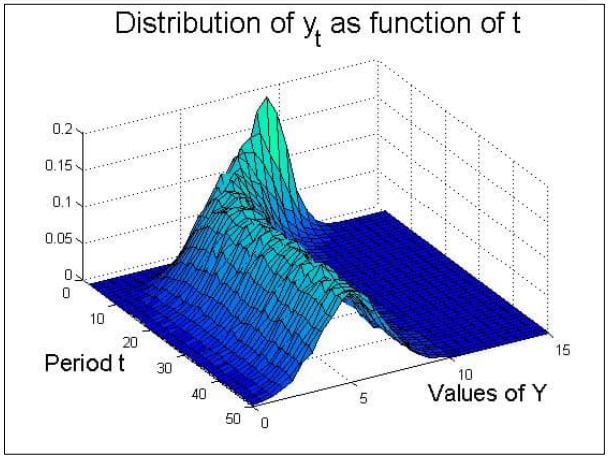
Stationarity of an AR(1) Process

If |b| < 1 there is a stationary process that satisfies

(see Appendix 2)

$$Y_t = a + bY_{t-1} + \varepsilon_t$$



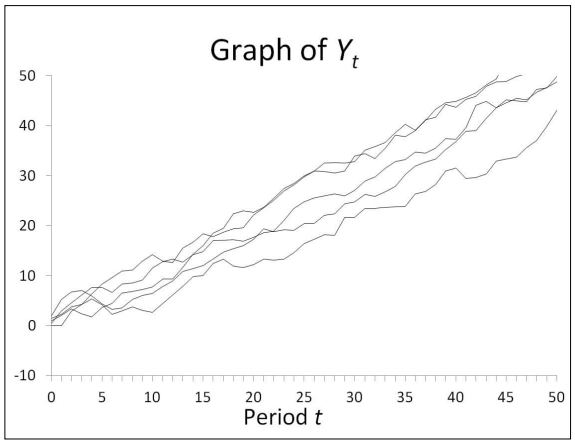


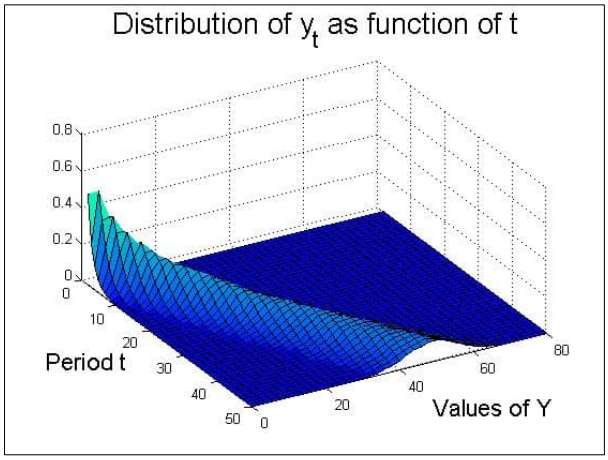
Non-stationary AR(1) Process

If $|b| \geq 1$ there is no stationary process that satisfies

$$Y_t = a + bY_{t-1} + \varepsilon_t$$

Example:





The Lag Operator

The lag operator $L(\cdot)$ transforms one series $\{Y_t\}_{-\infty}^{+\infty}$ into another by shifting the original series back by one period ("consider Y_{t-1} instead of Y_t")

$L(\cdot)$ is a linear operator that allows semi-algebraic operations:

$$L(ay_t + bx_t) = aL(y_t) + bL(x_t) = ay_{t-1} + bx_{t-1}$$

$$L^n y_t = L(L \ldots (L(Ly_t))) = y_{t-n}$$

$$(a + bL^m)(c + dL^n)y_t = (ac + bcL^m + adL^n + bdL^{m+n})y_t = \\ = acy_t + bcy_{t-m} + ady_{t-n} + bdy_{t-n-m}$$

If |b|<1 then $\left(\sum_{i=0}^{\infty} b^i L^i\right)(1 - bL)y_t = y_t$ (For details, see Appendix 3)

Stationarity of an AR(p) Process

Consider $y_t = a + b_1 y_{t-1} + b_2 y_{t-2} \ldots + b_p y_{t-p} + \varepsilon_t$

Using $L(\cdot)$ re-write this as

$$y_t = a + b_1 L y_t + b_2 L^2 y_t \ldots + b_p L^p y_t + \varepsilon_t$$

$$y_t - b_1 L y_t - b_2 L^2 y_t \ldots - b_p L^p y_t = a + \varepsilon_t$$

$$(1 - b_1 L - b_2 L^2 \ldots - b_p L^p) y_t = a + \varepsilon_t$$

Consider $\lambda_1, \lambda_2, \ldots \lambda_p$ (real/complex) such that:

$$(1 - b_1 L - b_2 L^2 \ldots - b_p L^p) = (1 - \lambda_1 L)(1 - \lambda_2 L) \ldots (1 - \lambda_p L)$$

Stationarity of an AR(p) Process

For any set of constant b_s', the coefficients λ_s' can *always* be found:

- either by solving the *characteristic equation*:

$$(1 - b_1 z - b_2 z^2 - \ldots - b_p z^p) = 0$$

and then inverting the solutions z_i^*:

$$\lambda_1 = \frac{1}{z_1^*} \quad \lambda_2 = \frac{1}{z_2^*} \quad \ldots \quad \lambda_p = \frac{1}{z_p^*}$$

- or directly by solving the polynomial:

$$(\lambda^p - b_1 \lambda^{p-1} - b_2 \lambda^{p-2} - \ldots - b_p) = 0$$

Stationarity of an AR(p) Process

An $AR(p)$ process is stationary if all the p solutions (z_i^*) to

$$(1 - b_1 z - b_2 z^2 - \ldots - b_p z^p) = 0$$

lie *outside* the unit (imaginary) circle (i.e., are greater than one in absolute value if real, or modulus if complex).

Equivalently, an AR(p) process is stationary if all the p solutions to

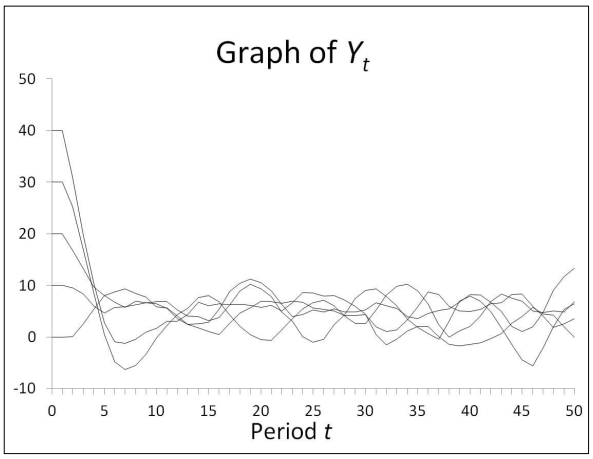
$$(\lambda^p - b_1 \lambda^{p-1} - b_2 \lambda^{p-2} - \ldots - b_p) = 0$$

$\lambda_i = \frac{1}{z_i^*}$ **are smaller than one in absolute value or in modulus**

Example of stationary AR(2) Process

Consider $Y_t = 1 + 1.5Y_{t-1} - 0.7Y_{t-2} + \varepsilon_t$

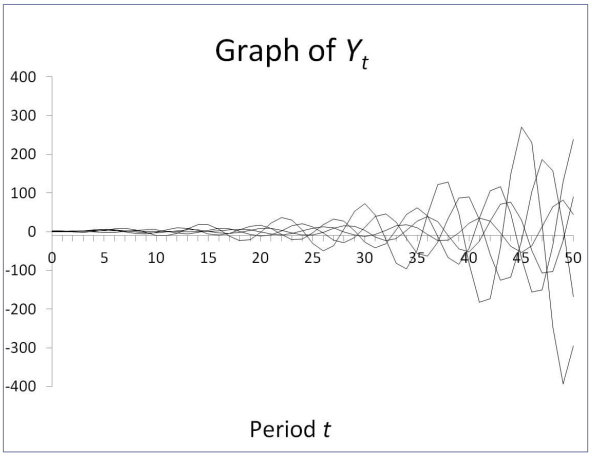
For this process $\lambda_{1,2} = \frac{1.5}{2} \pm i\frac{\sqrt{0.55}}{2}$ with modulus $\sqrt{0.7} < 1$.



Example of non-stationary AR(2) Process

Consider $Y_t = 1 + 1.5Y_{t-1} - 1.2Y_{t-2} + \varepsilon_t$

For this process $\lambda_{1,2} = \frac{1.5}{2} \pm i\frac{\sqrt{2.55}}{2}$ with modulus $\sqrt{1.2} > 1$.



Stationarity of a MA(q) Process

A MA process is *always* stationary: the mean and the variance of the process do not depend on t…

$$E[y_t] = \mu + \underbrace{E[\varepsilon_t] + \phi_1 E[\varepsilon_{t-1}] + \ldots + \phi_q E[\varepsilon_{t-q}]}_{=0} = \mu$$

$$Var[y_t] = Var[\varepsilon_t] + Var[\phi_1\varepsilon_{t-1}] + \ldots + Var[\phi_q\varepsilon_{t-q}] \ldots + \underbrace{\sum_{i\neq j} 2\phi_i\phi_j Cov[\varepsilon_i, \varepsilon_j]}_{=0}$$
$$= \sigma^2 + \phi_1^2\sigma^2 + \ldots + \phi_q^2\sigma^2$$

$$COV[y_t, y_{t-q}] = \phi_q Var(\varepsilon_{t-q}) = \phi_q\sigma^2$$
$$COV[y_t, y_{t-1}] = \phi_1 Var(\varepsilon_{t-q}) = \phi_1\sigma^2 + \phi_2{}^2\sigma^2 + \ldots + \phi_q{}^2\sigma^2$$

ARIMA Models

A stochastic process "y" is integrated of order "d" if we need to differentiate "d" times "y" to generate a stationary variable.

$$y_t \sim I[d] \Longrightarrow \Delta^d y_t \sim I[0]$$

where

$$\Delta = 1 - L$$

is the difference operator: $\Delta y_t = y_t - y_{t-1}$

A stochastic process admits an $ARIMA(p, d, q)$ representation if it is integrated of order d and its d-difference is a stationary $ARMA(p, q)$ process.

Part IV. Correlogram

- Autocorrelation and partial autocorrelation
- AR processes
- MA processes
- ARMA processes

Autocovariance and Autocorrelation

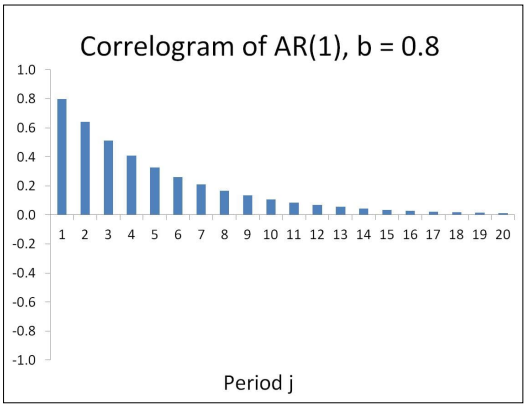
Define the autocovariance $\gamma_j = E[(y_t - \mu)(y_{t-j} - \mu)]$

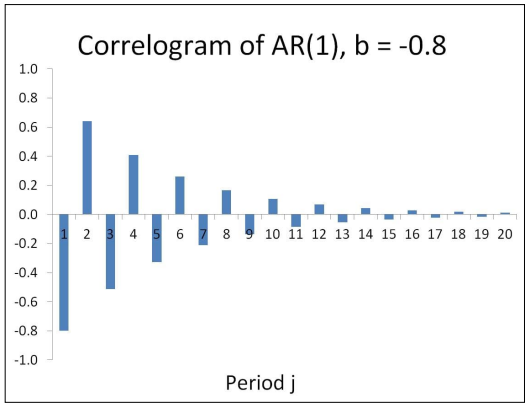
Define the autocorrelation $\rho_j = \frac{\gamma_j}{\gamma_0}$

The autocorrelation function (ACF or correlogram) is the graph of the autocorrelation against j.

Correlogram of an AR(1) Process

Recall the AR(1) process $y_t = a + by_{t-1} + \varepsilon_t$

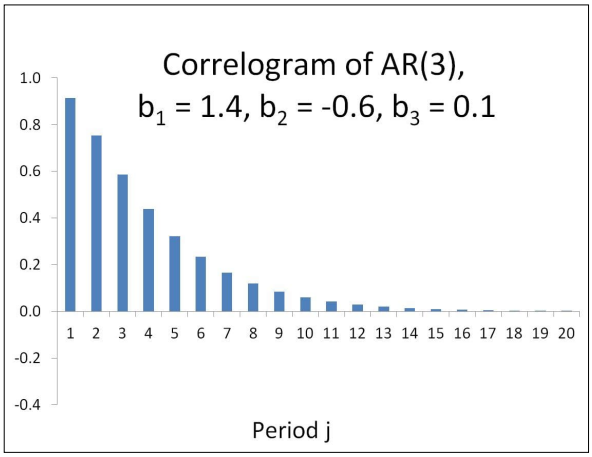


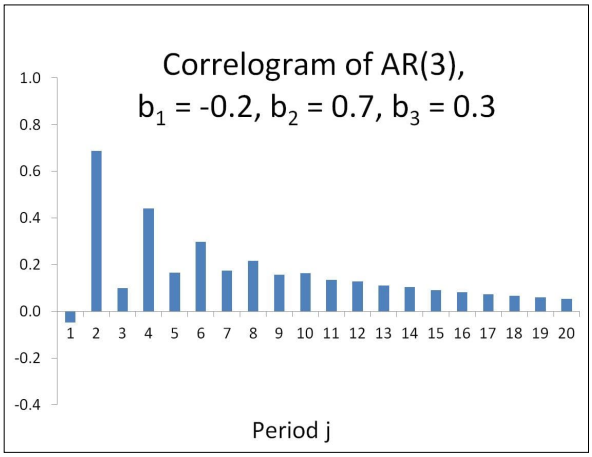


For a stationary $AR(1)$ process $\rho_j = b\rho_{j-1} \Rightarrow \rho_j = b^j$

The correlogram follows the deterministic part of the process

Correlogram of an AR(3) Process





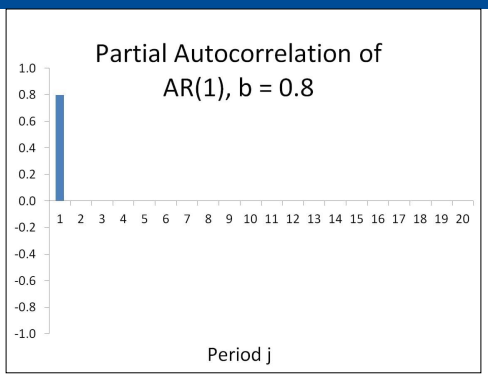
See Appendix 4 for further details.

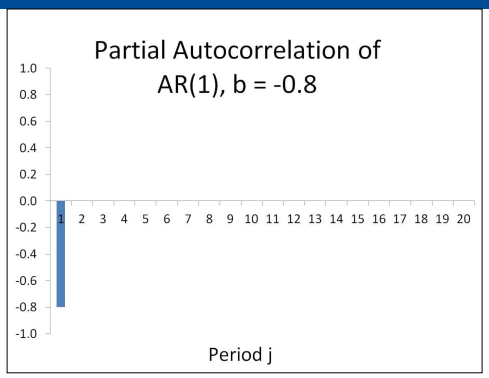
Partial-autocorrelation Function

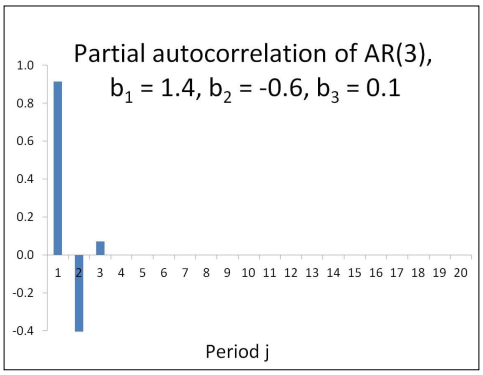
The pth partial autocorrelation is the last coefficient of a linear regression of y_t on its p immediate lags, that is the effect of y_{t-p} on y_t after controlling for y_{t-1}, y_{t-2}… and y_{t-p+1}

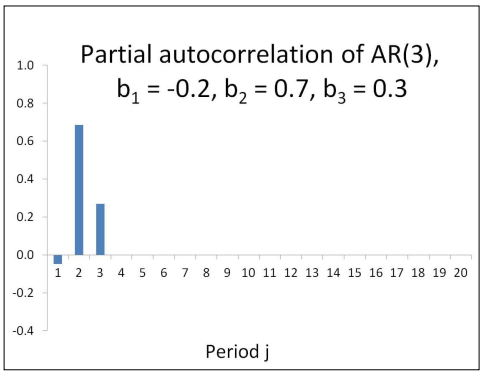
What would be the $p + j$ *partial* autocorrelation for an $AR(p)$ process?

PACF of an AR(1) and AR(3) Process









Autocovariance of a MA(q) Process

Consider $Cov[y_t, y_{t-j}]$ for a $MA(q)$ process.

Notice that y_t and y_{t-j} only have max[q-j+1,0] ε's in common:

$$y_t - \mu = \varepsilon_t + \phi_1\varepsilon_{t-1} + \phi_2\varepsilon_{t-2} + \phi_3\varepsilon_{t-3} + \ldots + \phi_q\varepsilon_{t-q}$$

$$y_{t-2} - \mu = \varepsilon_{t-2} + \phi_1\varepsilon_{t-3} + \ldots + \phi_{q-2}\varepsilon_{t-q} + \phi_{q-1}\varepsilon_{t-q-1} + \phi_q\varepsilon_{t-q-2}$$

$$Cov[y_t, y_{t-j}] = 0 \text{ if } j > q$$

The autocovariance and the autocorrelation of an $MA(q)$ process drop to zero after q lags.

Partial autocorrelation of a MA(q) Process

Consider a MA(1) process. Consider regressing y_t on y_{t-1} and y_{t-2}

$$y_t - \mu = \varepsilon_t + \phi_1 \varepsilon_{t-1}$$

$$y_{t-1} - \mu = \varepsilon_{t-1} + \phi_1 \varepsilon_{t-2}$$

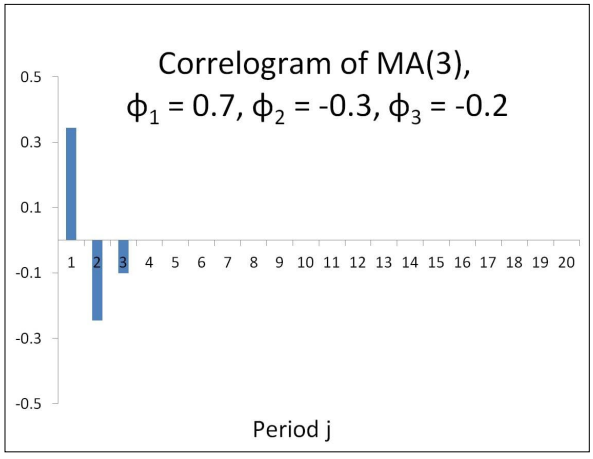
$$y_{t-2} - \mu = \varepsilon_{t-2} + \phi_1 \varepsilon_{t-3}$$

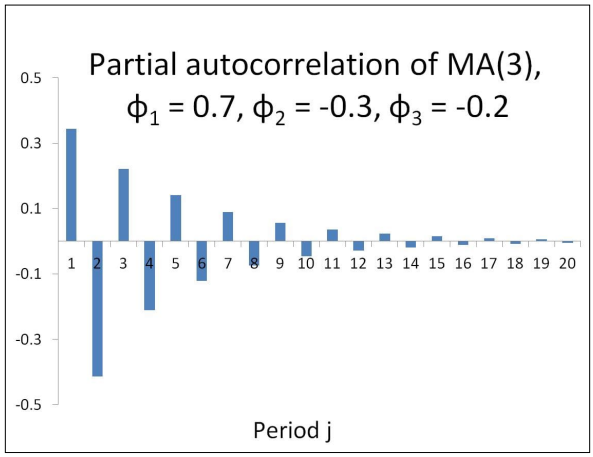
Because $Cov[y_{t-j}, y_{t-j-1}] \neq 0$ the coefficients on y_{t-1}, y_{t-2} is not zero.

The *j*-partial autocorrelation of an $MA(q)$ process is different from zero for all values of *j*.

ACF and PACF for MA(q) Process

The ACF and PACF of an MA process display opposite features to those of an AR process

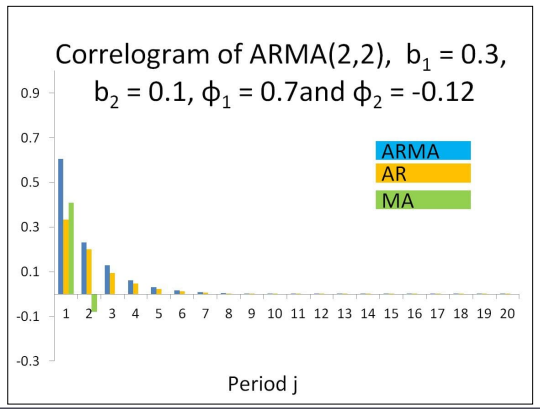


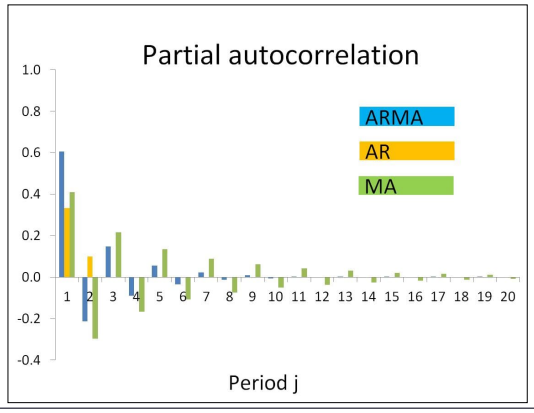


ACF and PACF of an ARMA(p,q) Process

The autocorrelation and partial autocorrelation are a mixture of the $AR(p)$ and $MA(q)$:

- the autocorrelation is non-harmonic for q lags (MA), decays geometrically after q lags (AR)
- the partial autocorrelation behaves non-harmonically for the first p periods (AR), and decay after period p (MA).





Part V. Box-Jenkins Methodology

- Identification of candidate ARIMA models
- Estimation and model selection
- Diagnostic checks

Box-Jenkins Methodology

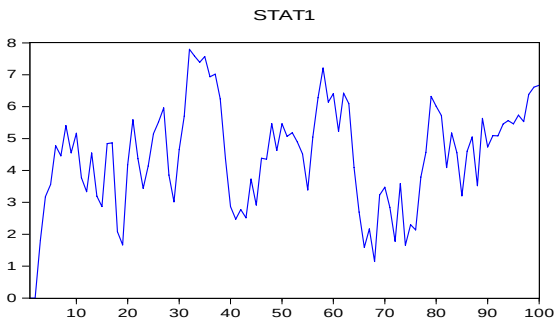
The Box-Jenkins methodology is a strategy to select a model that best replicates the statistical properties of an observed time series.

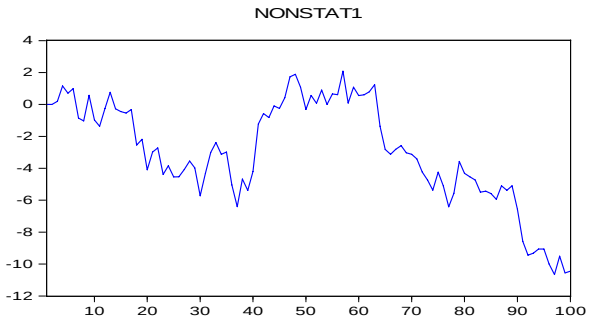
Steps:

1. Identification: use the estimated ACF and PACF to indentify candidate models
2. Estimation and selection: estimate and compare candidate models, and select one
3. Diagnostics: ensure that the residuals of the model mimic a white-noise

Step 1.1 identification, plotting the series

Does the series look stationary? If not, how?





Step 1.2: using the ACF and the PACF

To decide on the model, consider that:

- White noise: ACF and PACF zero
- $MA(q)$: ACF drops to zero after q lags
- $AR(p)$: PACF drops to zero after p lags
- $ARMA(p,q)$: the ACF and PACF of an ARMA is different from zero at all lags (see figures in the appendix)

Strategy: start with a parsimonious low order AR model, obtain the residuals and check the ACF and PACF of the residuals to decide on the MA component.

Step 2: estimation and model evaluation

Most software packages estimate the coefficients of the model, provide diagnostics for significance of coefficients, and statistics for goodness of fit.

Decide on number of lags and evaluate the model using:

- Schwarz Bayesian Criterion (SBC or BIC or SIC)

$$SBC = T\ln(SSR) + (p + q + 1)\ln(T)$$

- Akaike Information Criterion (AIC)

$$AIC = T\ln(SSR) + 2(p + q + 1)$$

- *SSR*: Sum of Squared Residuals; *T*: # of usable observations
- Both criteria strike the balance between **better fit** and **model parsimony**. The SBC "prefers" parsimonious models. AIC favors forecasting performance.

Step 2: estimation and model evaluation

Example: for a simulated series with true model

$$y_t = 1.6y_{t-1} - 0.63y_{t-2} + \varepsilon_t + 1.6\varepsilon_{t-1} - 0.63\varepsilon_{t-2}$$

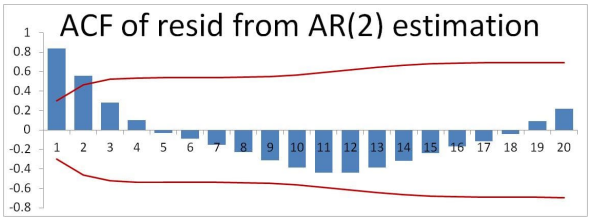
one would choose AR(3) based on AIC and SBC below:

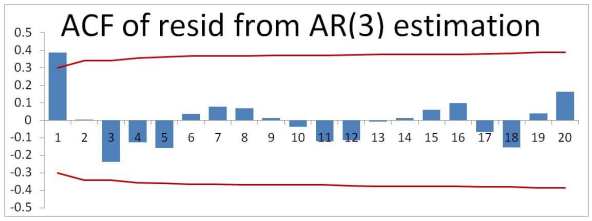
	AR(1)	AR(2)	AR(3)
b1	1.001***	2.007***	2.845***
b2		-1.011***	-2.708***
b3			0.863***
AIC	8.479	4.492	3.255
SBC	8.505	4.545	3.335

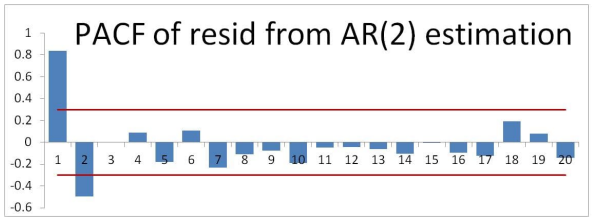
Let's also consider AR(2) as a candidate…

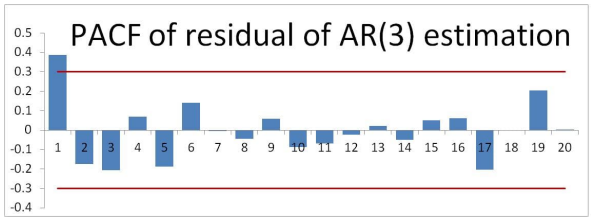
Step 3: diagnostics

For the selected model(s) check if residuals follow a white-noise process. Plot ACF and PACF of residuals, and test for autocorrelation up to j to be zero (Q statistic)







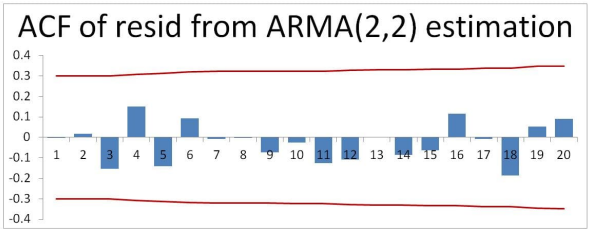


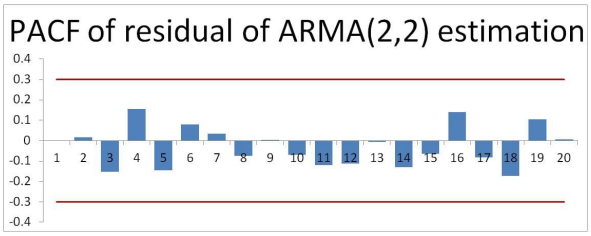
Residuals are not white noise: AR(2) residuals show 2 statistically significant lags and AR(3) show significant autocorrelation at lag 1.

Putting all steps together

Estimate AR(2) and AR(3) including MA(2)

	AR(2)	AR(3)
b1	1.978***	2.012***
b2	-0.98***	-1.050***
b3		0.03
ma(1)	1.265***	1.230***
ma(2)	-0.489***	-0.458***
AIC	3.020	3.255
SBC	3.127	3.335





Under the ARMA(2,2) estimation all coefficients are significantly different from zero and the residuals are not serially correlated. The AIC and SBC also select ARMA(2,2) over the alternative.

Part VI. Forecasting

- Forecasting with AR models
- Forecast error variance

The j-step ahead forecast of an AR(2) model

Given an AR model, one can construct the j-step ahead conditional expectation of y_{t+j} by iterating the difference equation implied by the true model.

$$y_t = a + b_1 y_{t-1} + b_2 y_{t-2} + \varepsilon_t$$

$$E[y_t | y_{t-1}, y_{t-2}] = a + b_1 y_{t-1} + b_2 y_{t-2} + E[\varepsilon_t | y_{t-1}, y_{t-2}]$$

$$y_{t+1} = a + b_1 y_t + b_2 y_{t-1} + \varepsilon_{t+1}$$

$$E[y_{t+1} | y_{t-1}, y_{t-2}] = a + b_1 E[y_t | y_{t-1}, y_{t-2}] + b_2 y_{t-1}$$

$$y_{t+j} = a + b_1 y_{t+j-1} + b_2 y_{t+j-2} + \varepsilon_{t+j}$$

$$E[y_{t+j} | y_{t-1}, y_{t-2}] = a + b_1 E[y_{t+j-1} | y_{t-1}, y_{t-2}] + b_2 E[y_{t+j-2} | y_{t-1}, y_{t-2}]$$

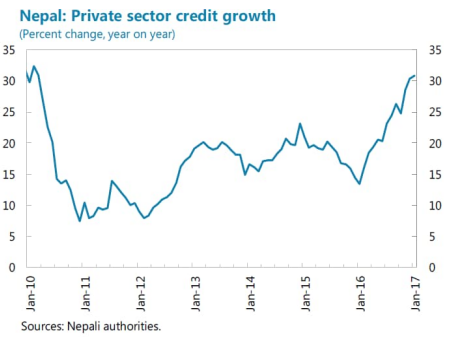
Convergence

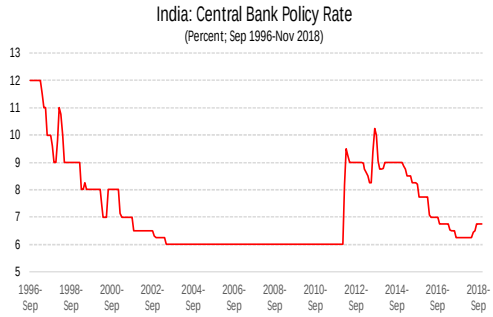
The forecast (conditional expected value) approaches the deterministic component of the AR process. Eventually

$$\lim_{j \to \infty} E[y_{t+j} | y_t, y_{t-1} \cdots y_{t-p}] = \mu$$

If the parameters of the model are estimated consistently, for large T the forecast error would follow the same AR process as Y. Eventually the variance of the forecast error would converge to the variance of the process (see Appendix 6).

Back to our first questions: How to model univariate time series?





- Univariate models may seem overly simplified, but they are useful:
  - 1. Essential to build intuition and clarify concepts in time series analysis.
  - 2. Often univariate models give as good short-term forecasts as more complicated multivariate models.
  - 3. Often it is straightforward to generalize univariate models to vectors of variables.
- What about other factors besides past values of the same time series? (Nepal: GDP growth? India: structural break?)

Conclusion

If a *stationary* process is believed to describe properly the dynamic and statistical properties of a variable Y then:

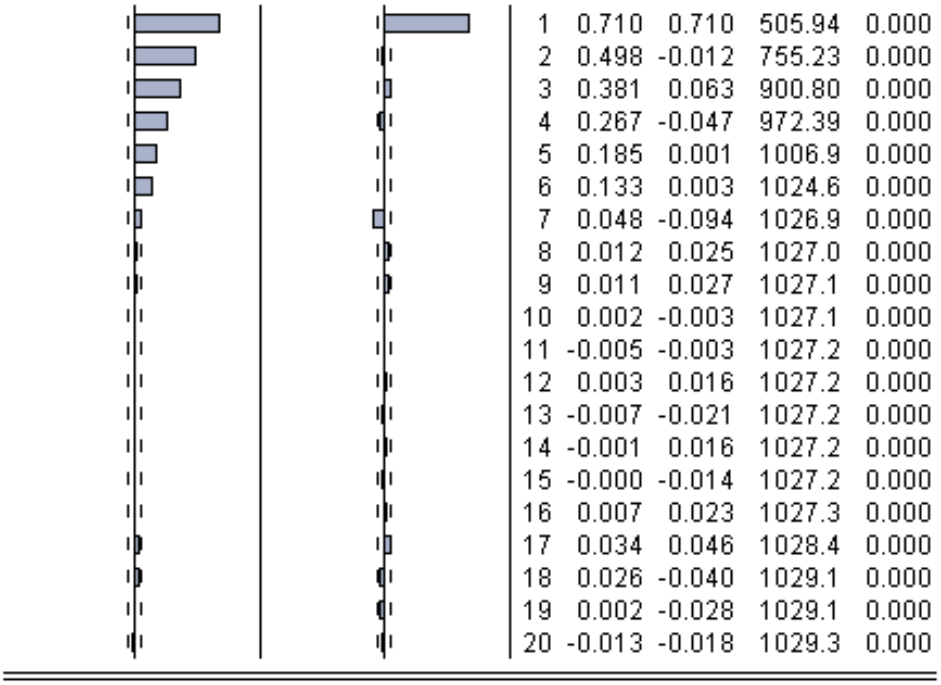
- an observed time series $\{y_t\}_{t=0}^{T}$ is enough to make simple inferences about mean, variance, autoregressive or moving average coefficients of Y
- Box-Jenkins methodology helps identify the appropriate model to describe Y. This methodology prefers simple models to complicated models
- The estimated model can be used, by forward iteration, to make forecasts on Y

Appendix 0. ACF and PACF

ACF and PACF for White Noise

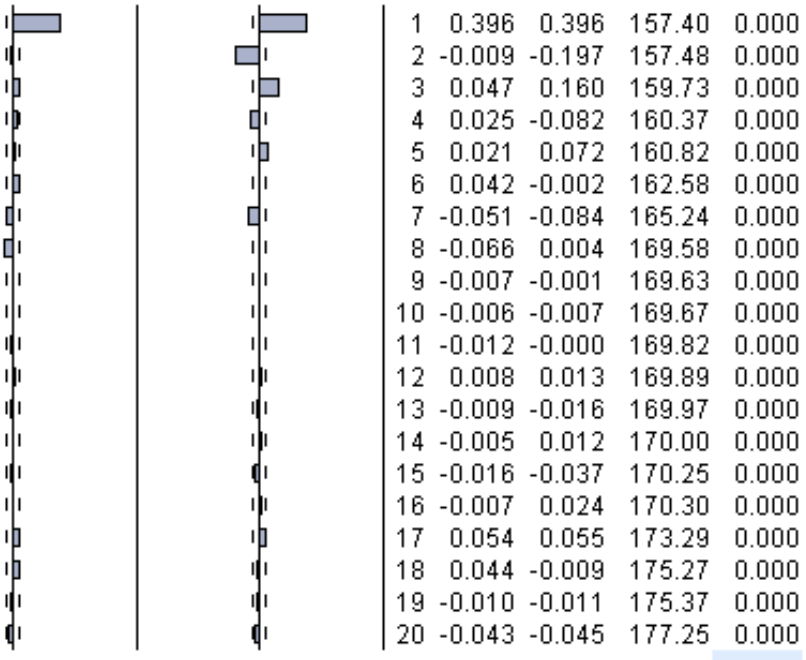
Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.020	0.020	0.3940	0.530
		2	-0.043	-0.043	2.2189	0.330
		3	0.064	0.066	6.3316	0.097
		4	0.003	-0.002	6.3410	0.175
		5	-0.007	-0.002	6.3946	0.270
		6	0.069	0.065	11.142	0.084
		7	-0.060	-0.065	14.802	0.039
		8	-0.048	-0.039	17.101	0.029
		9	0.012	0.001	17.256	0.045
		10	-0.003	0.001	17.263	0.069
		11	-0.021	-0.014	17.714	0.088
		12	0.026	0.021	18.387	0.104
		13	-0.022	-0.017	18.861	0.127
		14	0.007	0.014	18.913	0.168
		15	-0.007	-0.019	18.969	0.215
		16	-0.026	-0.024	19.636	0.237
		17	0.053	0.057	22.545	0.165
		18	0.028	0.020	23.356	0.177
		19	-0.012	-0.002	23.500	0.216
		20	-0.023	-0.029	24.042	0.241

ACF and PACF for AR(1)



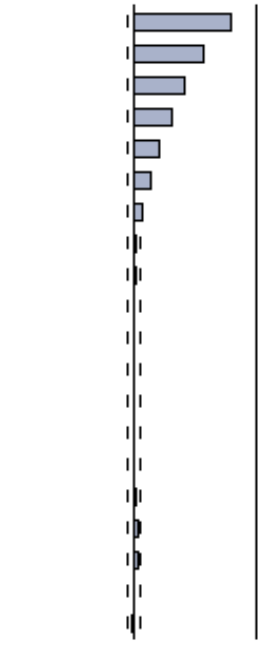
Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.710	0.710	505.94	0.000
		2	0.498	-0.012	755.23	0.000
		3	0.381	0.063	900.80	0.000
		4	0.267	-0.047	972.39	0.000
		5	0.185	0.001	1006.9	0.000
		6	0.133	0.003	1024.6	0.000
		7	0.048	-0.094	1026.9	0.000
		8	0.012	0.025	1027.0	0.000
		9	0.011	0.027	1027.1	0.000
		10	0.002	-0.003	1027.1	0.000
		11	-0.005	-0.003	1027.2	0.000
		12	0.003	0.016	1027.2	0.000
		13	-0.007	-0.021	1027.2	0.000
		14	-0.001	0.016	1027.2	0.000
		15	-0.000	-0.014	1027.2	0.000
		16	0.007	0.023	1027.3	0.000
		17	0.034	0.046	1028.4	0.000
		18	0.026	-0.040	1029.1	0.000
		19	0.002	-0.028	1029.1	0.000
		20	-0.013	-0.018	1029.3	0.000

ACF and PACF for MA(1)



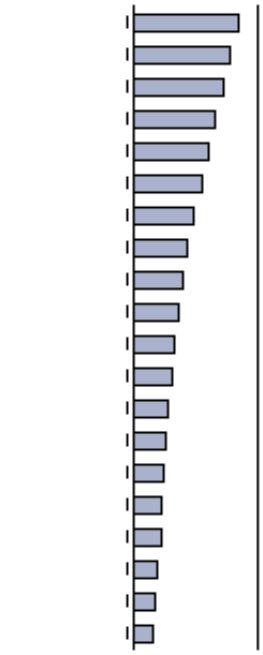
Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.396	0.396	157.40	0.000
		2	-0.009	-0.197	157.48	0.000
		3	0.047	0.160	159.73	0.000
		4	0.025	-0.082	160.37	0.000
		5	0.021	0.072	160.82	0.000
		6	0.042	-0.002	162.58	0.000
		7	-0.051	-0.084	165.24	0.000
		8	-0.066	0.004	169.58	0.000
		9	-0.007	-0.001	169.63	0.000
		10	-0.006	-0.007	169.67	0.000
		11	-0.012	-0.000	169.82	0.000
		12	0.008	0.013	169.89	0.000
		13	-0.009	-0.016	169.97	0.000
		14	-0.005	0.012	170.00	0.000
		15	-0.016	-0.037	170.25	0.000
		16	-0.007	0.024	170.30	0.000
		17	0.054	0.055	173.29	0.000
		18	0.044	-0.009	175.27	0.000
		19	-0.010	-0.011	175.37	0.000
		20	-0.043	-0.045	177.25	0.000

ACF and PACF for ARMA(1,1)



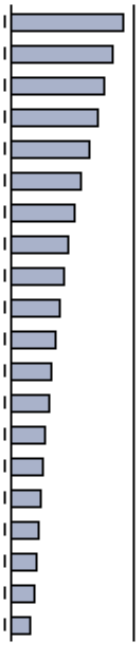
	AC	PAC	Q-Stat	Prob
1	0.835	0.835	699.29	0.000
2	0.596	-0.334	1055.9	0.000
3	0.438	0.197	1248.7	0.000
4	0.315	-0.142	1348.2	0.000
5	0.220	0.072	1396.9	0.000
6	0.144	-0.080	1417.9	0.000
7	0.067	-0.042	1422.4	0.000
8	0.022	0.061	1422.9	0.000
9	0.010	-0.007	1423.0	0.000
10	0.002	-0.002	1423.0	0.000
11	-0.002	0.006	1423.0	0.000
12	-0.001	0.002	1423.0	0.000
13	-0.004	-0.014	1423.0	0.000
14	-0.003	0.016	1423.1	0.000
15	0.001	-0.009	1423.1	0.000
16	0.013	0.051	1423.2	0.000
17	0.030	-0.000	1424.1	0.000
18	0.026	-0.055	1424.8	0.000
19	0.005	-0.009	1424.8	0.000
20	-0.011	-0.006	1425.0	0.000

ACF and PACF for ARMA(2,1)



	AC	PAC	Q-Stat	Prob
1	0.880	0.880	776.88	0.000
2	0.813	0.169	1440.1	0.000
3	0.755	0.054	2012.8	0.000
4	0.687	-0.051	2487.6	0.000
5	0.626	-0.018	2882.1	0.000
6	0.574	0.011	3214.1	0.000
7	0.506	-0.083	3472.8	0.000
8	0.459	0.028	3685.6	0.000
9	0.423	0.041	3866.1	0.000
10	0.385	0.006	4015.9	0.000
11	0.350	-0.006	4140.2	0.000
12	0.325	0.019	4247.2	0.000
13	0.293	-0.020	4334.5	0.000
14	0.272	0.017	4409.8	0.000
15	0.250	-0.007	4473.6	0.000
16	0.232	0.011	4528.2	0.000
17	0.222	0.039	4578.5	0.000
18	0.201	-0.042	4619.9	0.000
19	0.178	-0.032	4652.3	0.000
20	0.159	-0.013	4678.1	0.000

ACF and PACF for ARMA(2,2)



		AC	PAC	Q-Stat	Prob
1		0.945	0.945	896.65	0.000
2		0.863	-0.288	1645.2	0.000
3		0.797	0.186	2284.1	0.000
4		0.730	-0.168	2819.5	0.000
5		0.665	0.087	3265.4	0.000
6		0.604	-0.092	3633.2	0.000
7		0.541	-0.009	3929.0	0.000
8		0.489	0.057	4170.3	0.000
9		0.447	0.011	4372.4	0.000
10		0.408	-0.003	4541.0	0.000
11		0.373	0.008	4681.9	0.000
12		0.342	0.003	4800.8	0.000
13		0.313	-0.012	4900.3	0.000
14		0.288	0.020	4984.7	0.000
15		0.266	-0.012	5056.6	0.000
16		0.248	0.041	5119.1	0.000
17		0.233	-0.011	5174.2	0.000
18		0.213	-0.054	5220.4	0.000
19		0.190	-0.008	5257.1	0.000
20		0.169	0.000	5286.4	0.000

Appendix 1. Stationarity

AR(p) process: interpretation of stationarity

Consider $y_t = a + b_1 y_{t-1} + b_2 y_{t-2} + \ldots + b_p y_{t-p} + \varepsilon_t$

$$E[y_t] = a + b_1 E[y_{t-1}] + b_2 E[y_{t-2}] + \ldots + b_p E[y_{t-p}] + E[\varepsilon_t]$$

By definition of stationarity $E[y_t] = E[y_{t-1}] = \mu$ $\quad E[\varepsilon_t] = 0$

and

$$\mu = a + b_1 \mu + b_2 \mu + \ldots + b_p \mu$$

Substituting $a = \mu - b_1 \mu - b_2 \mu - \ldots - b_p \mu$ into the AR(p):

$$(y_t - \mu) = b_1 (y_{t-1} - \mu) + b_2 (y_{t-2} - \mu) + \ldots + b_p (y_{t-p} - \mu) + \varepsilon_t$$

The mean μ can be interpreted as the "equilibrium" value of *Y*. Deviations from μ are attenuated each period according to the same dynamics that drives Y_t

Appendix 2. MA(∞) representation of an AR(1) process

MA representation of an AR(1) process

Notice that

$$
\begin{aligned}
Y_t &= a + bY_{t-1} + \varepsilon_t \\
&= a + b[a + bY_{t-2} + \varepsilon_{t-1}] + \varepsilon_t \\
&= \ldots \\
&= \sum_{i=0}^{\infty} b^i a + \sum_{i=0}^{\infty} b^i \varepsilon_{t-i}
\end{aligned}
$$

If $|b| < 1$

$$Y_t = \frac{a}{1-b} + \sum_{i=0}^{\infty} b^i \varepsilon_{t-i}$$

which is a MA(∞) process.

Appendix 3. Lag operators

Definition: lag operator

The lag operator transforms one series $\{y_t\}_{-\infty}^{+\infty}$ into another by shifting the original series back by one period:

$Ly_t = y_{t-1}$ applying the lag operator to y_t means "consider y_{t-1} instead of y_t"

The following seemingly algebraic operations mean:

$L^2 y_t = L(Ly_t) = y_{t-2}$

$L^n y_t = L \ldots (Ly_t) = y_{t-n}$

$bLy_t = b(Ly_t) = by_{t-1}$

$Lc = c$

$L(y_t + x_t) = L(y_t) + L(x_t) = y_{t-1} + x_{t-1}$

$(a + bL^n)y_t = ay_t + bL^n y_t = ay_t + by_{t-n}$

$$(a + bL^m)(c + dL^n)y_t = (a + bL^m)(cy_t + dL^n y_t) = (a + bL^m)(cy_t + dy_{t-n})$$
$$= (a + bL^m)cy_t + (a + bL^m)dy_{t-n}$$
$$= acy_t + bcy_{t-m} + ady_{t-n} + bdy_{t-n-m}$$

Example

Consider $y_t = \{5 \quad 6 \quad 7 \quad 8 \quad 10\}$ and $x_t = \{0 \quad 1 \quad 2 \quad 3 \quad 4\}$

Ly_t $\quad y_{t-1} = \{n.a. \quad 5 \quad 6 \quad 7 \quad 8\}$

$L^2 y_t = L(Ly_t)$ $\quad y_{t-2} = \{n.a. \quad n.a. \quad 5 \quad 6 \quad 7\}$

$bLy_t = b(Ly_t) = by_{t-1}$ $\quad by_{t-1} = \{n.a. \quad 5b \quad 6b \quad 7b \quad 8b\}$

$L(y_t + x_t) = L(y_t) + L(x_t) = y_{t-1} + x_{t-1}$

$y_{t-1} + x_{t-1} = \{n.a. \quad 5 \quad 7 \quad 9 \quad 11\}$

Definition: Inverse of the lag operator

Consider now applying the following operators to $\{y_t\}_{-\infty}^{+\infty}$

$$\left(1 + bL + b^2L^2 + b^3L^3 + \cdots + b^kL^k\right)(1 - bL)y_t$$

$$y_t + by_{t-1} + b^2y_{t-2} + b^3y_{t-3} + \ldots + b^ky_{t-k} +$$
$$\ldots - by_{t-1} - b^2y_{t-2} - b^3y_{t-3} - \ldots - b^{k+1}y_{t-k-1} = y_t - b^{k+1}y_{t-k-1}$$

If $|b| < 1$ and the series is bounded, then as k increases the term $b^{k+1}y_{t-k-1}$ becomes negligible. One can define $(1 - bL)^{-1}$ so that:

$$(1 - bL)^{-1}(1 - bL)y_t = y_t$$

$$(1 - bL)^{-1} \equiv \sum_{i=0}^{\infty} b^iL^i$$

Definition: factorization of lag operators

For any set of constant *b*s', there *always* exist *real* or *imaginary* number λs' such that:

$$(1 - b_1L - b_2L^2 - \ldots - b_pL^p) = (1 - \lambda_1L)(1 - \lambda_2L)\ldots(1 - \lambda_pL)$$

These can be found:

- either by solving the *characteristic equation*:

$$(1 - b_1z - b_2z^2 - \ldots - b_pz^p) = 0$$

and then inverting the solutions z_i^*:

$$\lambda_1 = \frac{1}{z_1^*} \quad \lambda_2 = \frac{1}{z_2^*} \quad \ldots \quad \lambda_p = \frac{1}{z_p^*}$$

- by solving the polynomial:

$$(\lambda^p - b_1\lambda^{p-1} - b_2\lambda^{p-2} - \ldots - b_p) = 0$$

Appendix 4. Derivation of AR(p) autocovariances

Stationary AR(p) process: autocorrelation (I)

To compute the autocovariance consider:

$$(y_t - \mu) = b_1(y_{t-1} - \mu) + b_2(y_{t-2} - \mu) + \ldots + b_p(y_{t-p} - \mu) + \varepsilon_t$$

and recall that $E[\varepsilon_t, (y_{t-j} - \mu)] = 0$ for all $j > 0$. Hence:

$$\gamma_0 = E[(y_t - \mu)^2] = b_1 E[(y_{t-1} - \mu)(y_t - \mu)] \ldots$$
$$+ b_2 E[(y_{t-2} - \mu)(y_t - \mu)] + b_p E[(y_{t-p} - \mu)(y_t - \mu)] + E[\varepsilon_t, (y_t - \mu)]$$

$$\ldots$$

$$\gamma_0 = b_1\gamma_1 + b_2\gamma_2 + \ldots + b_p\gamma_p + \sigma_\varepsilon^2$$

$$\gamma_1 = E[(y_t - \mu)(y_{t-1} - \mu)] = b_1 E[(y_{t-1} - \mu)(y_{t-1} - \mu)] \ldots$$
$$+ b_2 E[(y_{t-2} - \mu)(y_{t-1} - \mu)] + b_p E[(y_{t-p} - \mu)(y_{t-1} - \mu)]$$

$$\ldots$$

$$\gamma_1 = b_1\gamma_0 + b_2\gamma_1 + \ldots + b_p\gamma_{p-1}$$

Stationary AR(p) process: autocorrelation (II)

$$\gamma_2 = E[(y_t - \mu)(y_{t-2} - \mu)] = b_1 E[(y_{t-1} - \mu)(y_{t-2} - \mu)] \ldots$$
$$+ b_2 E[(y_{t-2} - \mu)(y_{t-2} - \mu)] + b_p E[(y_{t-p} - \mu)(y_{t-2} - \mu)]$$

...

$$\gamma_2 = b_1\gamma_1 + b_2\gamma_0 + \ldots + b_p\gamma_{p-2}$$

$$\gamma_p = E[(y_t - \mu)(y_{t-p} - \mu)] = b_1 E[(y_{t-1} - \mu)(y_{t-p} - \mu)] \ldots$$
$$+ b_2 E[(y_{t-2} - \mu)(y_{t-p} - \mu)] + b_p E[(y_{t-p} - \mu)(y_{t-p} - \mu)]$$

$$\gamma_p = b_1\gamma_{p-1} + b_2\gamma_{p-2} + \ldots + b_p\gamma_0$$

$$\gamma_j = E[(y_t - \mu)(y_{t-j} - \mu)] = b_1 E[(y_{t-1} - \mu)(y_{t-j} - \mu)] \ldots$$
$$+ b_2 E[(y_{t-2} - \mu)(y_{t-j} - \mu)] + b_p E[(y_{t-j} - \mu)(y_{t-j} - \mu)]$$

...

$$\gamma_j = b_1\gamma_{j-1} + b_2\gamma_{j-2} + \ldots + b_p\gamma_{j-p}$$

Stationary AR(p) process: autocorrelation (III)

To conclude:

$$\gamma_0 = b_1\gamma_1 + b_2\gamma_2 + \ldots + b_p\gamma_p + \sigma_\varepsilon^2$$

$$\gamma_1 = b_1\gamma_0 + b_2\gamma_1 + \ldots + b_p\gamma_{p-1}$$

$$\gamma_2 = b_1\gamma_1 + b_2\gamma_0 + \ldots + b_p\gamma_{p-2}$$

...

$$\gamma_p = b_1\gamma_{p-1} + b_2\gamma_{p-2} + \ldots + b_p\gamma_0$$

$$\gamma_j = b_1\gamma_{j-1} + b_2\gamma_{j-2} + \ldots + b_p\gamma_{j-p}$$

Appendix 5. Tests for autocorrelation

Tests for autocorrelation

One can test the hypothesis that the *j*th autocorrelation (partial autocorrelation) coefficient is equal to zero:

- For the autocorrelation, if true $\rho_j = 0$ (*MA(j-1)*) then

$$std(\hat{\rho}_j) = \sqrt{T^{-1}\left(1 + 2\sum_{i=1}^{j-1}\hat{\rho}_i^2\right)} \qquad pac_{jj} = 0$$

- For the partial autocorrelation, if true $pac_j = 0$ (AR(j-1)) then

$$std\left(\widehat{pac_j}\right) = \frac{1}{\sqrt{T}}$$

One can use these formulas to construct 95% confidence intervals around zero for the ACF and PACF.

Appendix 6. Forecasts

The j-step ahead forecast error of an AR(2)

If the parameters of the model were known, the forecast errors e_t would be exactly:

$$y_t - E[y_t|y_{t-1}, y_{t-2}] = (a + b_1 y_{t-1} + b_2 y_{t-2} + \varepsilon_t) - (a + b_1 y_{t-1} + b_2 y_{t-2})$$

$$e_t = \varepsilon_t$$

$$\begin{aligned} y_{t+1} - E[y_{t+1}|\dots] &= (a + b_1 y_t + b_2 y_{t-1} + \varepsilon_{t+1}) - (a + b_1 E[y_t|\dots] + b_2 y_{t-1}) \\ &= b_1 y_t - b_1 E[y_t|\dots] + \varepsilon_{t+1} \end{aligned}$$

$$e_{t+1} = b_1 e_t + \varepsilon_{t+1} = b_1 \varepsilon_t + \varepsilon_{t+1}$$

$$y_{t+j} - E[y_{t+j}|\dots] = (a + b_1 y_{t+j-1} + b_2 y_{t+j-2} + \varepsilon_{t+j}) - (a + b_1 E[y_{t+j-1}|\dots] + b_2 E[y_{t+j-2}|\dots])$$

$$e_{t+j} = b_1 e_{t+j-1} + b_2 e_{t+j-2} + \varepsilon_{t+j}$$

The prediction error follows the same AR(2) process as y.

Convergence of Var(e) to Var(Y)

What are the dynamic properties of the forecast error?

$$E[e_t] = 0 \qquad E[e_{t+1}] = b_1 E[e_t] + E[\varepsilon_{t+1}] = 0 \qquad E[e_{t+j}] = 0$$

For any ARMA, the expected value of the forecasting error based on the conditional expectation is zero.

For any ARMA, because the forecasting error follows the same stationary process as the ARMA, eventually the variance of the forecast error converges to the variance of the process.

Error due to parameter estimation

The forecasting error is:

$$y_t - E[y_t | y_{t-1}, y_{t-2}] = \varepsilon_t$$

Provided that the parameters of the model are known. If these are estimated, the forecasting error depends also on the error in estimating the coefficient

$$y_t - E[y_t | y_{t-1}, y_{t-2}, \ldots, y_{t-T}] = \varepsilon_t + v(y_{t-1}, y_{t-2}, \ldots, y_{t-T})$$

The forecasting error keeps following the ARMA process:

$$\lim_{j \to \infty} v\,(y_{t-1}, y_{t-2}, \ldots, y_{t-T}) = 0$$

If parameters are estimated consistently and with large T, this error *v* is small to begin with, and it will vanish.